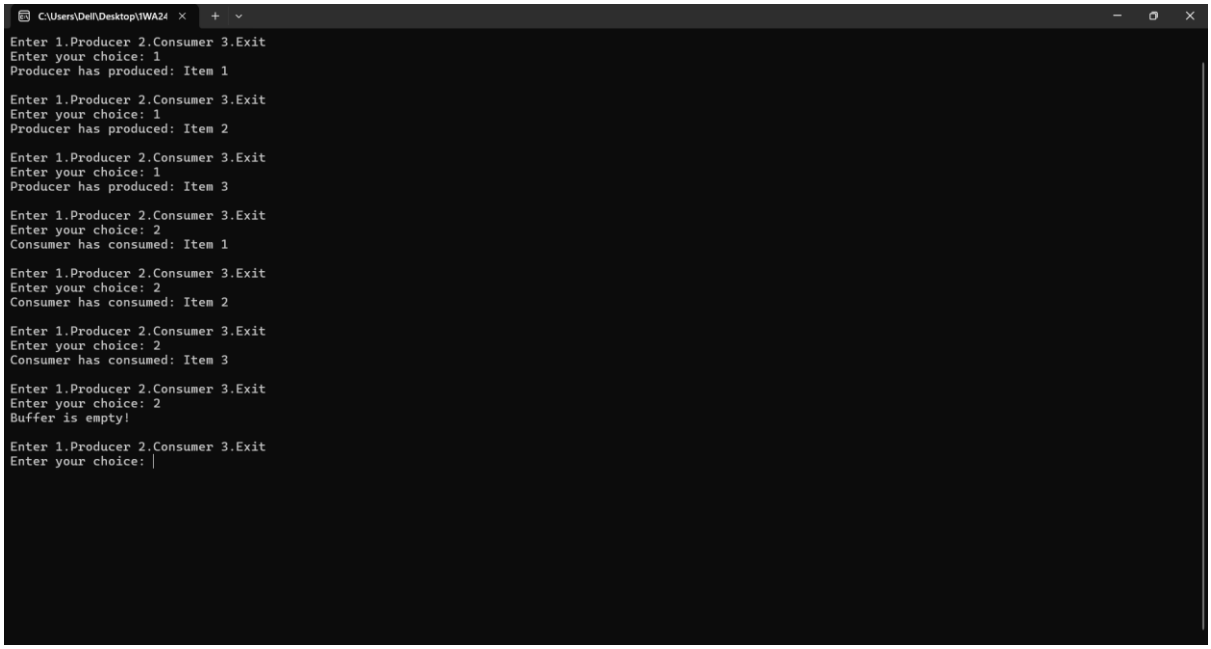
LAB 4

1)C program to simulate: Producer-Consumer problem using semaphores

```
#include <stdio.h>
#include <stdlib.h>
#define BUFFER_SIZE 5
int buffer[BUFFER_SIZE];
int in = 0, out = 0;
int count = 0;   // number of items in buffer
void produce(int item) {
    if (count == BUFFER_SIZE) {
        printf("Buffer is full!\n");
        return;
    }
    buffer[in] = item;
    printf("Producer has produced: Item %d\n", item);
    in = (in + 1) % BUFFER_SIZE;
    count++;
}
void consume() {
    if (count == 0) {
        printf("Buffer is empty!\n");
```

```
        return;
    }
    int item = buffer[out];
    printf("Consumer has consumed: Item %d\n", item);
    out = (out + 1) % BUFFER_SIZE;
    count--;
}
int main() {
    int choice;
    int item = 1;
    while (1) {
        printf("\nEnter 1.Producer 2.Consumer 3.Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);
        switch (choice) {
            case 1:
                produce(item++);
                break;
            case 2:
                consume();
                break;
            case 3:
                printf("Exiting...\n");
```

```
            exit(0);
        default:
            printf("Invalid choice!\n");
    }
  }
  return 0;
}
```



2)Write a C program to simulate: Dining-Philosopher's Problem

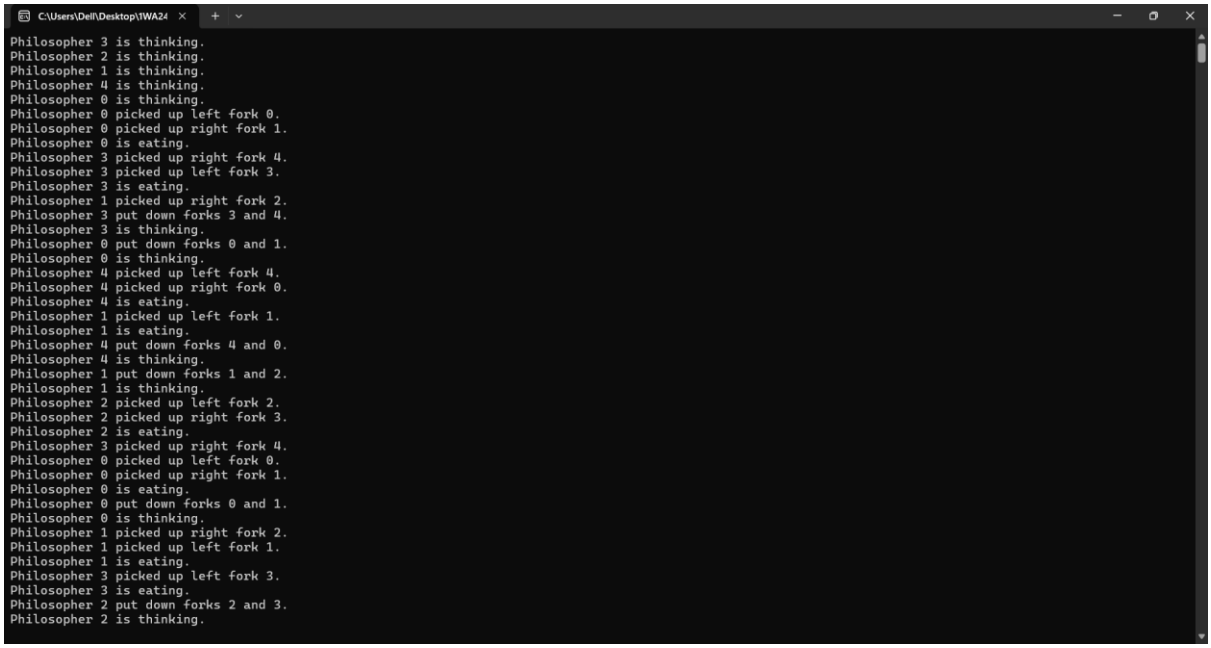
```
#include <stdio.h>
#include <pthread.h>
#include <unistd.h>
#define N 5
```

```
pthread_mutex_t forks[N];
pthread_t philosophers[N];
void* philosopher(void* num)
{
    int id = *(int*)num;
    int left = id;
    int right = (id + 1) % N;
    while (1)
    {
        printf("Philosopher %d is thinking.\n", id);
        sleep(1);
        if (id % 2 == 0)
        {
            pthread_mutex_lock(&forks[left]);
            printf("Philosopher %d picked up left fork %d.\n", id,
left);
            pthread_mutex_lock(&forks[right]);
            printf("Philosopher %d picked up right fork %d.\n", id,
right);
        }
        else
```

```
        {
            pthread_mutex_lock(&forks[right]);
            printf("Philosopher %d picked up right fork %d.\n", id,
right);
            pthread_mutex_lock(&forks[left]);
            printf("Philosopher %d picked up left fork %d.\n", id,
left);
        }
        printf("Philosopher %d is eating.\n", id);
        sleep(2);
        pthread_mutex_unlock(&forks[left]);
        pthread_mutex_unlock(&forks[right]);

        printf("Philosopher %d put down forks %d and %d.\n",
id, left, right);
    }
    return NULL;
}
int main()
{
    int i;
```

```
    int ids[N];
    for (i = 0; i < N; i++)
    {
        pthread_mutex_init(&forks[i], NULL);
        ids[i] = i;
    }
    for (i = 0; i < N; i++)
    {
        pthread_create(&philosophers[i], NULL, philosopher,
&ids[i]);
    }
    for (i = 0; i < N; i++)
    {
        pthread_join(philosophers[i], NULL);
    }
    for (i = 0; i < N; i++)
    {
        pthread_mutex_destroy(&forks[i]);
    }
    return 0;
}
```


C:\Users\Dell\Desktop\IWA24
Philosopher 3 is thinking.
Philosopher 2 is thinking.
Philosopher 1 is thinking.
Philosopher 4 is thinking.
Philosopher 0 is thinking.
Philosopher 0 picked up left fork 0.
Philosopher 0 picked up right fork 1.
Philosopher 0 is eating.
Philosopher 3 picked up right fork 4.
Philosopher 3 picked up left fork 3.
Philosopher 3 is eating.
Philosopher 1 picked up right fork 2.
Philosopher 3 put down forks 3 and 4.
Philosopher 3 is thinking.
Philosopher 0 put down forks 0 and 1.
Philosopher 0 is thinking.
Philosopher 4 picked up left fork 4.
Philosopher 4 picked up right fork 0.
Philosopher 4 is eating.
Philosopher 1 picked up left fork 1.
Philosopher 1 is eating.
Philosopher 4 put down forks 4 and 0.
Philosopher 4 is thinking.
Philosopher 1 put down forks 1 and 2.
Philosopher 1 is thinking.
Philosopher 2 picked up left fork 2.
Philosopher 2 picked up right fork 3.
Philosopher 2 is eating.
Philosopher 3 picked up right fork 4.
Philosopher 0 picked up left fork 0.
Philosopher 0 picked up right fork 1.
Philosopher 0 is eating.
Philosopher 0 put down forks 0 and 1.
Philosopher 0 is thinking.
Philosopher 1 picked up right fork 2.
Philosopher 1 picked up left fork 1.
Philosopher 1 is eating.
Philosopher 3 picked up left fork 3.
Philosopher 3 is eating.
Philosopher 2 put down forks 2 and 3.
Philosopher 2 is thinking.